NETWORKING

Optically-Accelerated AI Clusters Enabled by Interoperable Multi-Vendor OCSs: Use Cases and Opportunities

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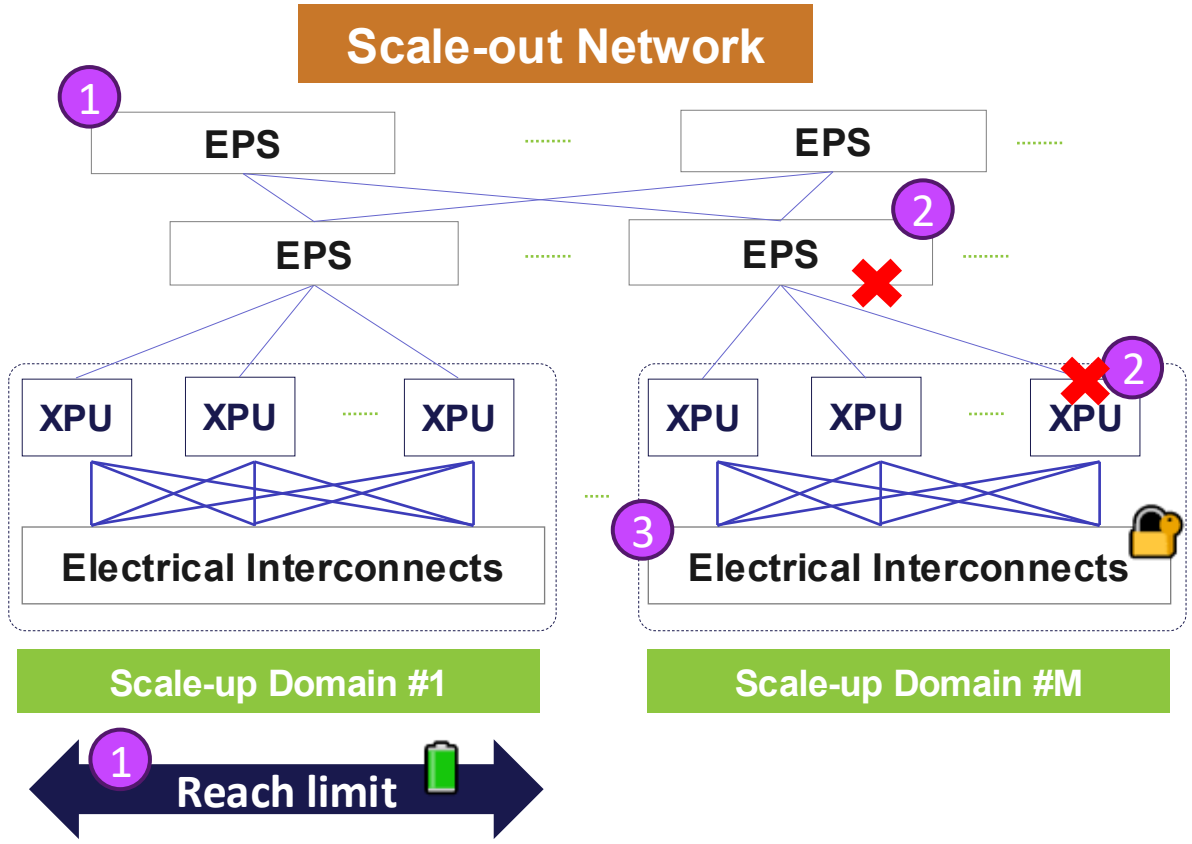
Table of Contents

1. Background & Challenges: Scaling AI Clusters
2. The Solution: Emerging optics technology and its application to AI Clusters
3. Optically-accelerated AI clusters (OAI) Architecture
4. Emerging Use Cases of OAI
    1. SDN-based Failure Recovery
    2. Workload optimization
5. Key Technical Challenges & Gaps
6. Ongoing Initiatives
    1. Bridging the Gaps at OCP OCS Subproject
    2. Multi-vendor Interoperability Demo at OFC2027 (OCP & IOWN Global Forum)
7. Call to Action

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Background & Challenges: Scaling AI Clusters

Exploding AI model sizes expose the critical limitations of traditional electrical interconnects



1 **Challenge 1 - The Bandwidth & Power Wall**

Scale-up (Reach Limit)
Scale-out (Power/Density Limit)

2 **Challenge 2 - Resiliency Bottlenecks**

Mean Time Between Failures (MTBF) drops dramatically as cluster scale grows.
(In 131,072 GPU cluster, the MTBF = 15 minutes)

3 **Challenge 3 - Vendor Lock-in**

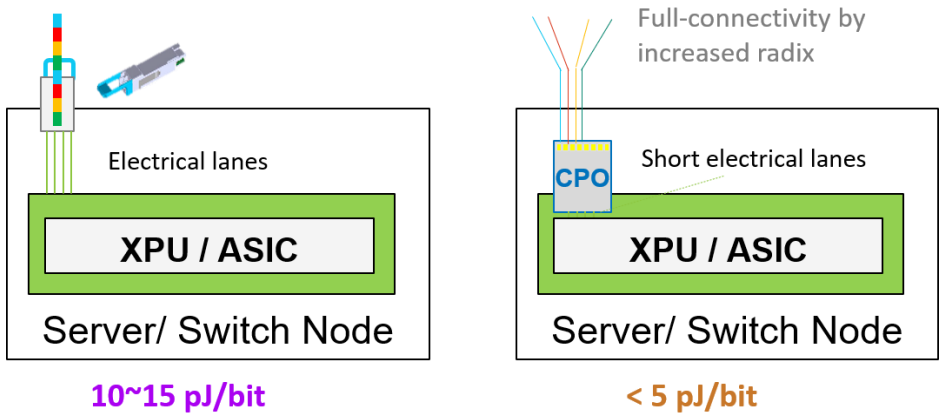
Proprietary scale-up systems
Lack of flexibility and drive up both CapEx and OpEx

Promising technologies: CPO / OCS

Co-Packaged Optics (CPO)

- Longer reach
- Increased radix with high port density
- Reduced energy consumption

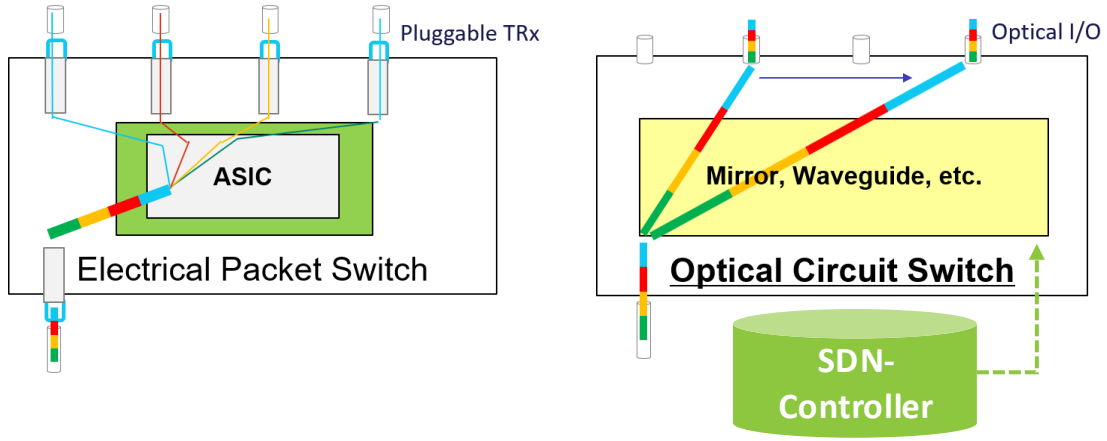
Current implementation vs CPO



Optical Circuit Switch (OCS)

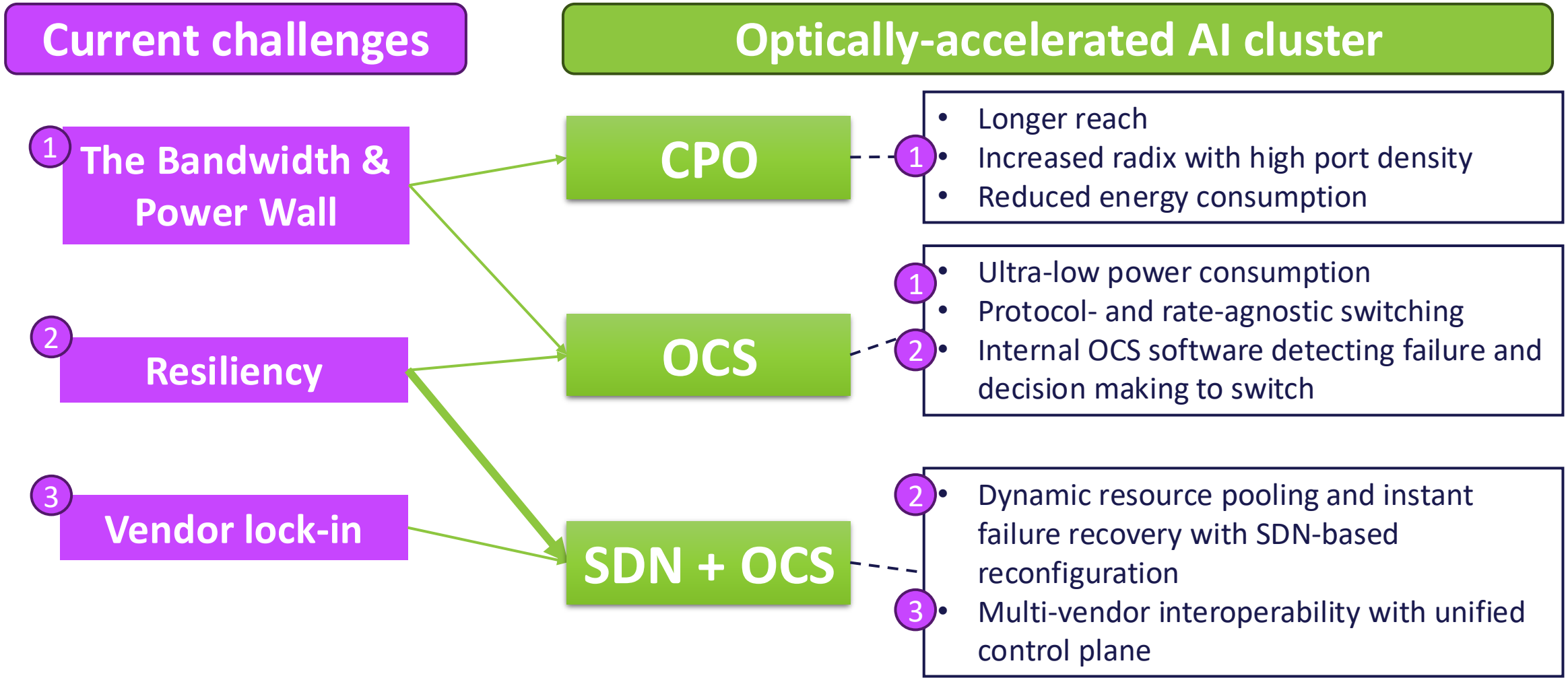
- Ultra-low power consumption
- L1 switching: Protocol- and rate-agnostic
- Reconfigurable physical connectivity with SDN

EPS vs OCS

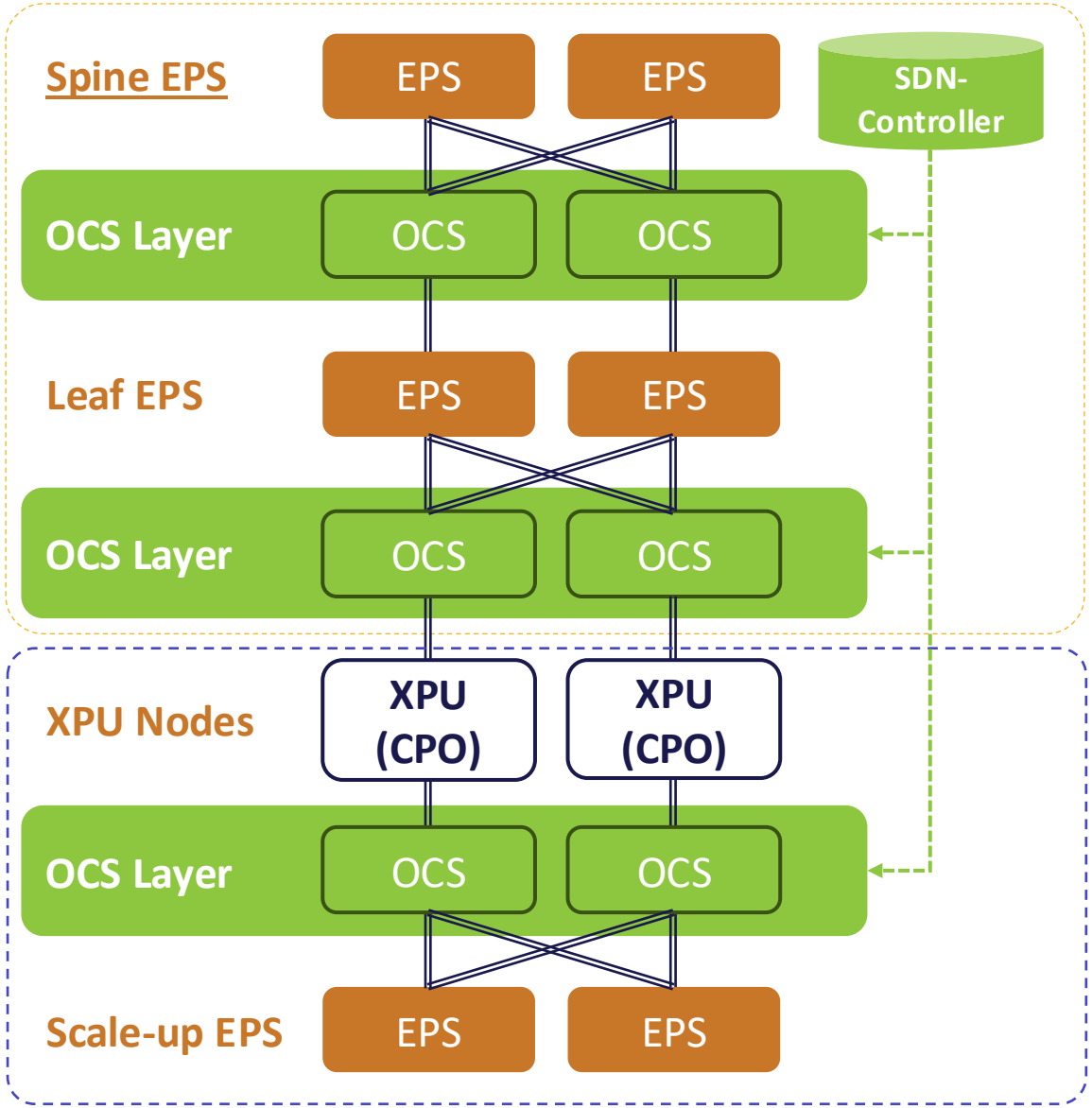


The Solution: Optically-accelerated AI clusters (OAI)

Design the future AI cluster architecture with CPO + OCS + SDN



The Architecture of OAI



Key idea

The Hybrid Foundation

By decoupling fine-grained logical routing (handled by **Electrical Packet Switches**) from coarse-grained physical layer connections (handled by **OCS**), unlocking unprecedented flexibility.

Dynamic Topology Reconfiguration via Programmable Physical Wiring

By coupling the **OCS layer** with an **SDN Controller**, the static physical connections become entirely programmable. This unlocks:

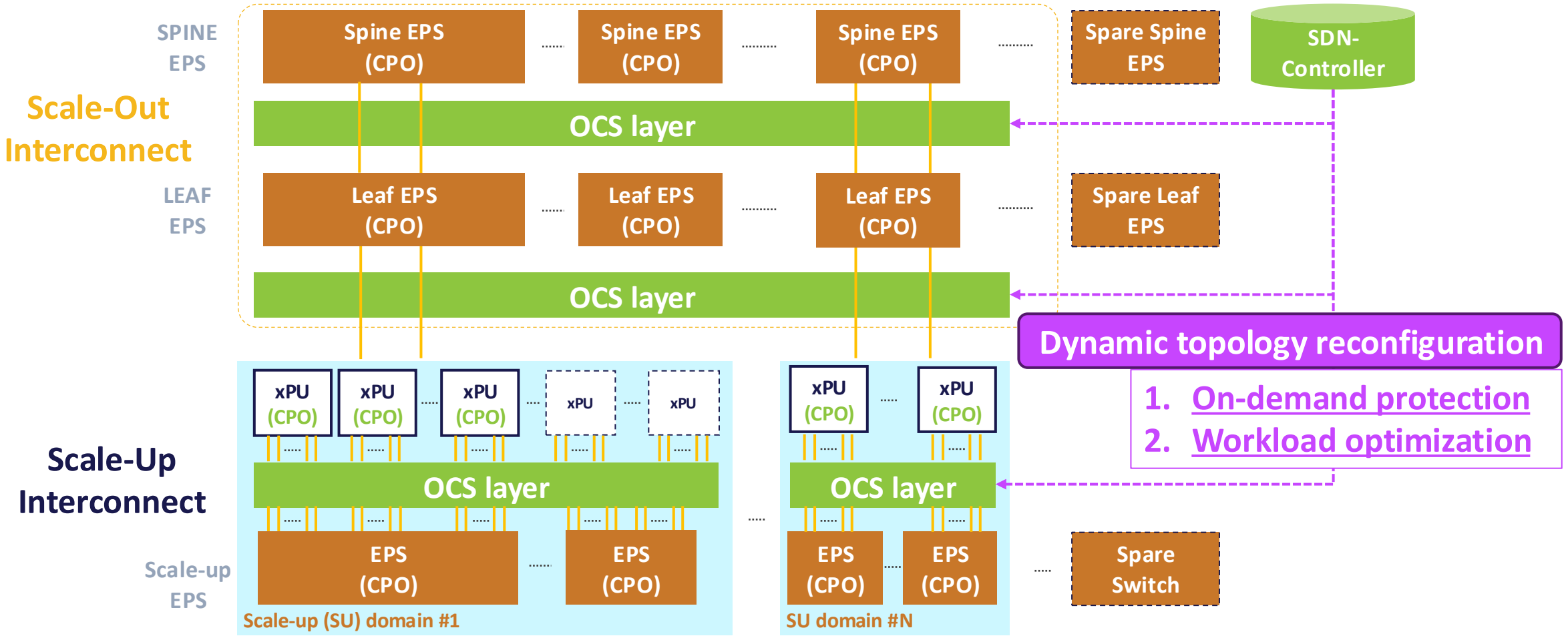
- **On-demand protection** (Automated failure bypass)
- **Workload optimization** (traffic-aware topology tuning)

Placement-Agnostic Flexibility

The core innovation functions regardless of where the OCS is deployed within the architecture. OCS can be installed with the Leaf, Spine, and **Scale-up**.

Versatile OCS Integration in the OAI Architecture

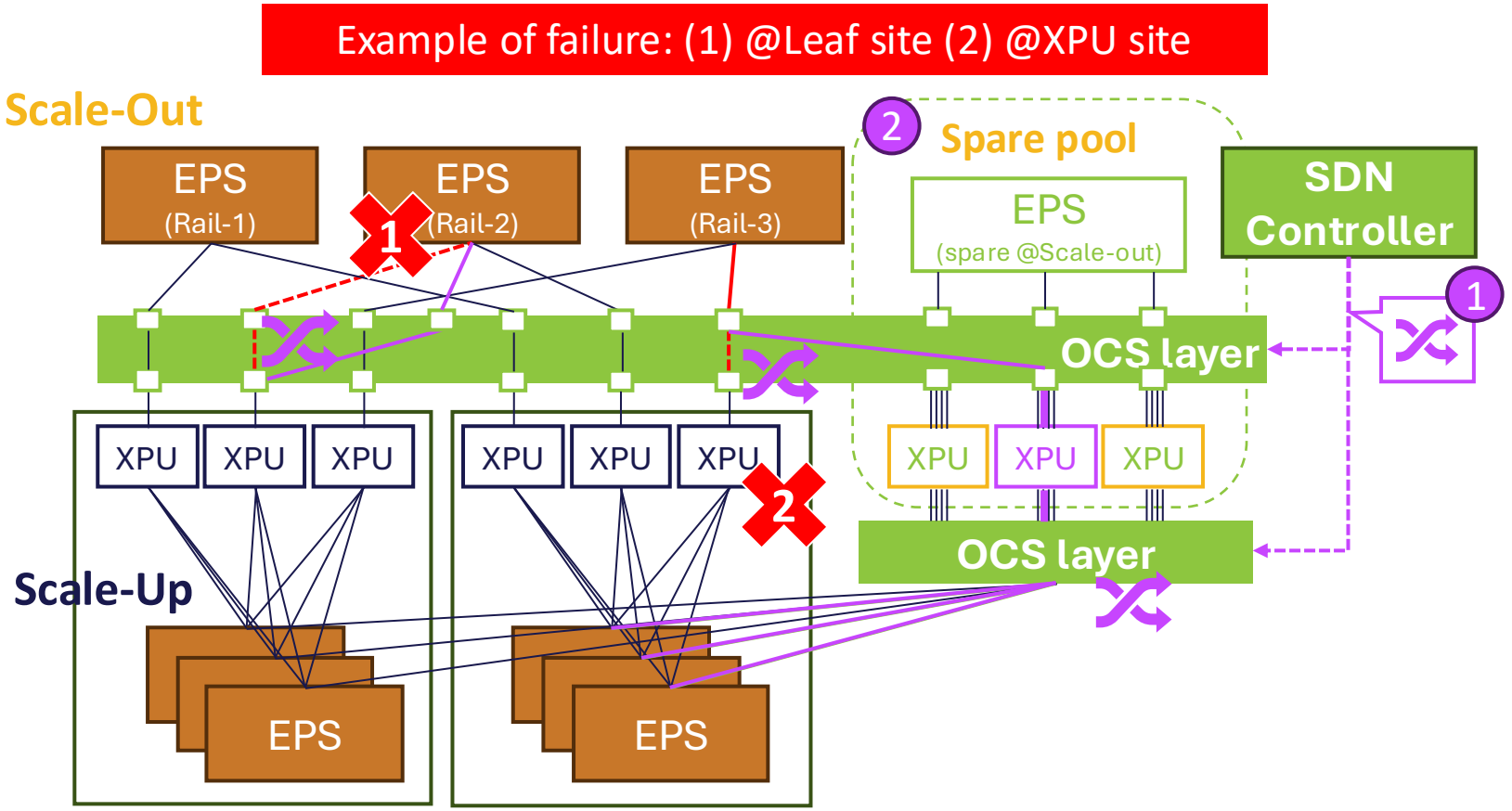
Unlocking multiple network capabilities by strategically deploying OCS layers alongside CPO-based electrical packet switches.



Use Case 1: SDN-based Failure Recovery

"From 1:1 Protection to Global Spare Pooling"

Minimize the cluster downtime with **dynamic reconfiguration** at intermediate OCS layer
Maximizing cluster reliability while minimizing CAPEX through **globally sharable spare resources pool**



Key innovations

1 **Dynamic reconfiguration**

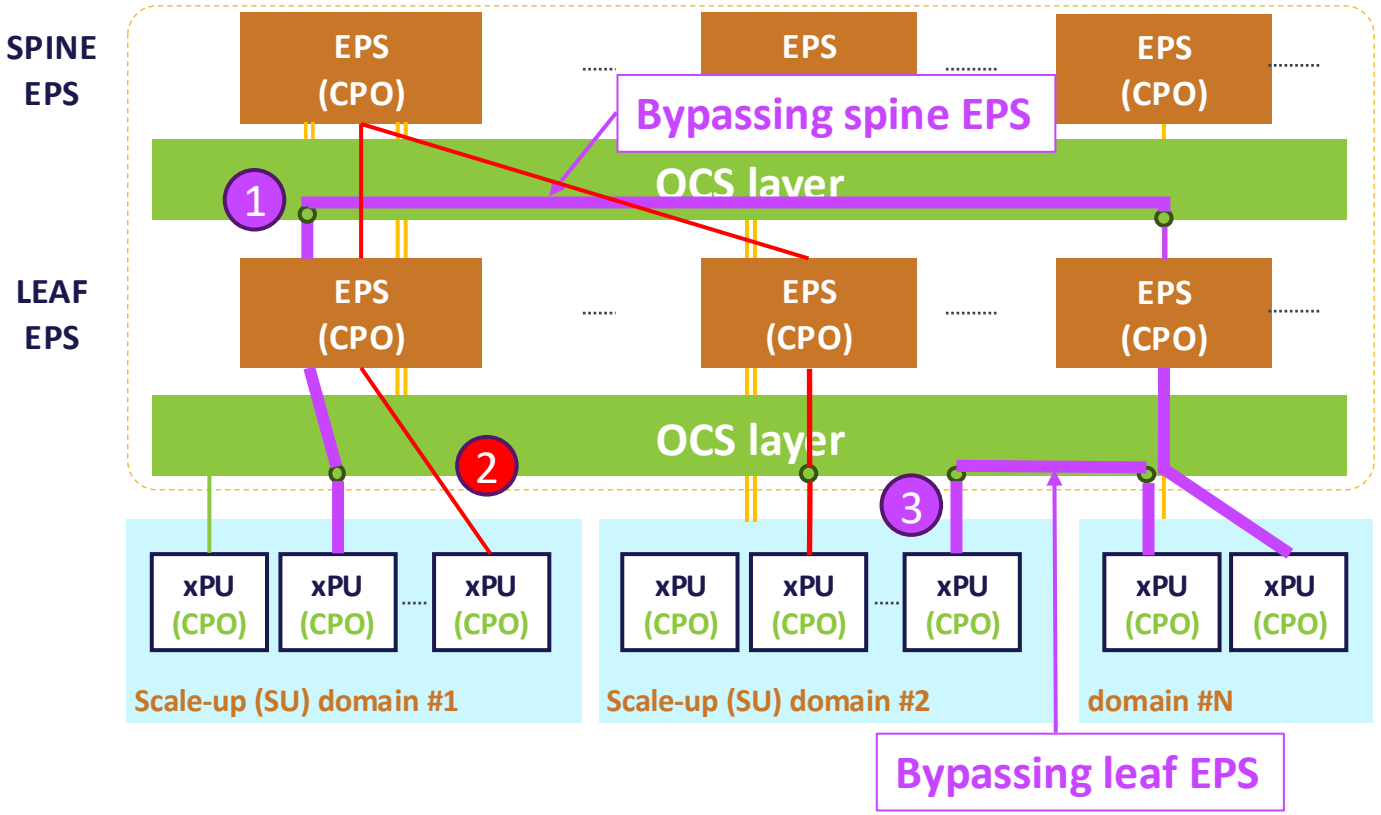
SDN controller can dynamically reroute optical paths around failed nodes to resume AI jobs

2 **Spare resource pool**

Original approach = 1:1 protection
Spare pooling = 1:N protection

Use Case 2: Optimizing AI workload with EPS & OCS-based direct connection

Reconfiguring physical connections on-demand to perfectly match specific AI workload communication patterns



Key innovations

1. Providing direct connectivity through OCS
3. (3) requires tight collaboration with AI job orchestrator
2. Coexistence of EPS & OCS communication

- AI job leverages specific collective communication, such as All-Reduce and All-to-All
- The SDN + OCS dynamically reconfigures optical connections to precisely match specific AI workload communication patterns
- Such topology adaptation minimizes latency and maximizes throughput, **significantly enhancing overall performance**

Key Technical Challenges & Gaps

SDN-based Failure Recovery

- *Challenge:* Relies on an **Unified Control Plane** and real-time telemetry / health monitoring capabilities.
- *Current Gaps*: standardized control plane for OCSs

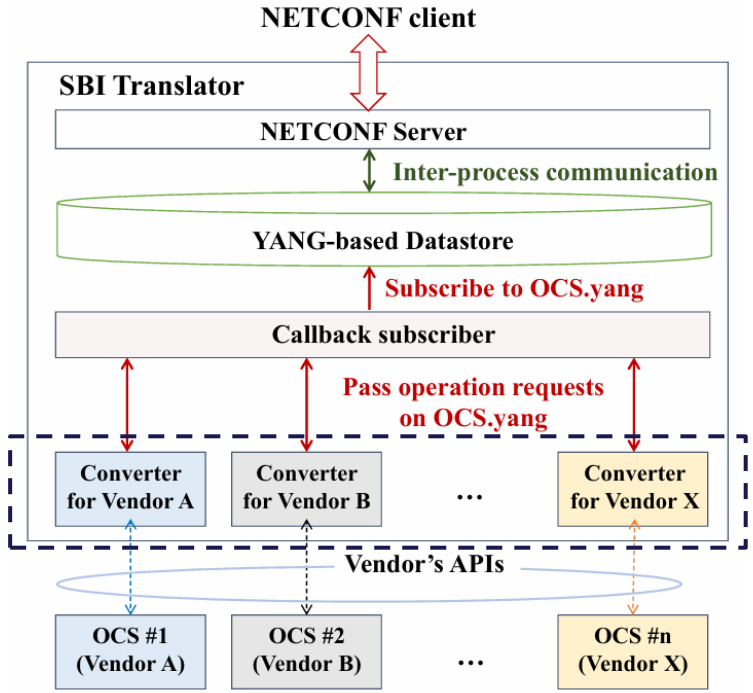
Optimizing AI workload

- *Challenge:* requires novel AI-OCS orchestration layer above the **Unified Control Plane**, built upon **real-time AI traffic visibility** and **hardware-software co-design**
- *Current Gaps*
  - AI traffic visibility: vital visibility into job phases and traffic patterns is lacking in standard deployments
  - HW/SW co-design for the optical switching gap: Optical reconfiguration introduces temporary physical disconnections which may disrupts RoCEv2 traffic. We need tight integration with collective communication libraries (e.g., NCCL) and SDN controller.

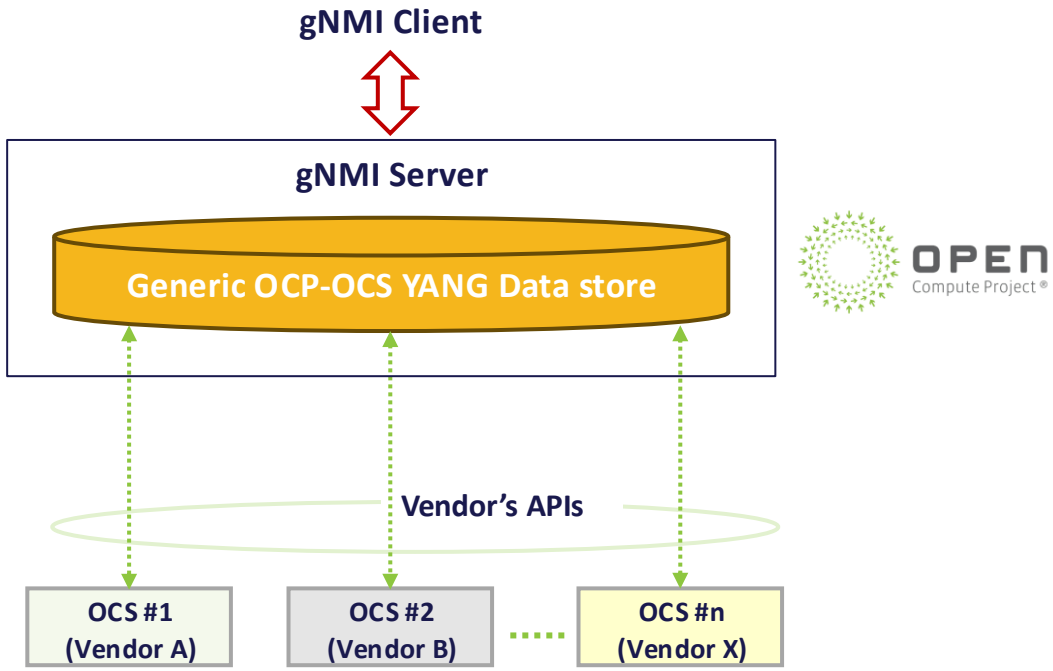
OCP-IOWN Global Forum Interop Demo

- Benefits of Generic OCP-OCS YANG Model
  - Open interface for Multi-Vendor Access Interoperability (Avoid Vendor Lock-In)
  - Much simpler SDN controller with direct access to any OCS Vendor without any SBI Translator (Converter for any specific Vendor's YANG model)

Multi-Vendor SDN Controller with SBI Translator



Multi-Vendor SDN Controller *without SBI Translator*

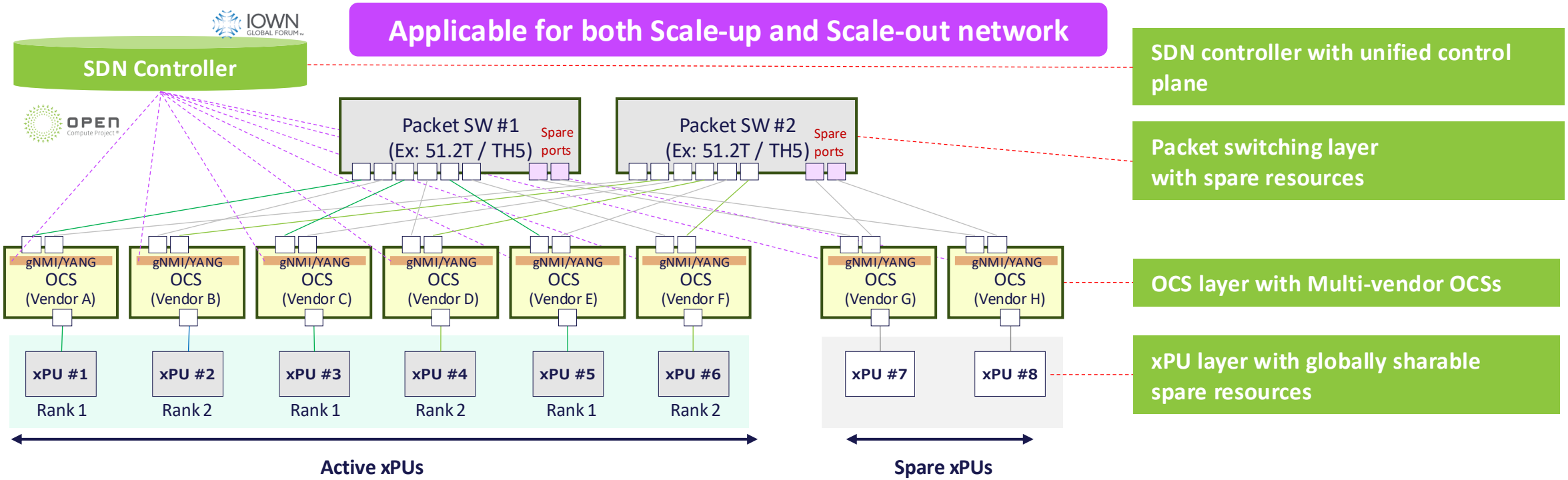


Anazawa, Kazuya, et al. "Experimental evaluation of an SDN controller for open optical-circuit-switched networks." Journal of Optical Communications and Networking 17.6 (2025): 498-513.

Overview of demonstration

The First Demonstration of Failure Recovery in OAI with Interoperable Multi-Vendor OCSs

1. **Live demonstration of interoperable multi-vendor OCS environments via OCP standard YANG model**
2. **Showcasing unified control plane capabilities in action through SDN controller provided by NTT**
3. **Validating feasibility of failure recovery with spare resource pool use-case discussed in IOWN-GF & OCP**



OCP-OCS Subproject - Activities



- **Bi-Weekly Meeting on every other Wednesday (9-10AM PST)**
  - Goal: Inform public about on going OCP-OCS Sub-Group working activities
  - Participants: OCS Vendors, Hyperscaler, Chip Vendors, Academia, etc..
  - Providing status update of three sub-group working committees
    - OCS SW steering working committee
    - OCS White Paper working committee
    - OCP-IOWN Global Forum Collaboration working Committee

- **OCS SW Steering Working Committee**
  - Goal: Standardized Open OCS
  - Participants: OCS Vendors and Hyperscaler
  - OCP-OCS Standard formation
    - OCP-OCS YANG Model R1.0 (Released)
    - OCP-OCS YANG Model R2.0 (Future Work)
    - Open-Source Centralize SDN Controller (Future Work)

- **OCS White Paper Working Committee**
  - Goal: Help educate public about OCS technologies, benefits and its high-level use cases based on available published papers
  - Participants: OCS Vendors, Optical System Vendor, enabling technology supplier vendor and Academia
  - List of white paper topics:
    - Introduction to OCS Technologies & Benefits (Published and summarized in OCP Online Training Academy)
    - Fast switching OCS, Technology and use cases (Being presented at this OCP Global Summit 2026)
    - OCS Operations and Management (Future discussion)

- **OCP-IOWN Global Forum Collaboration Working Committee**
  - Goal: Demonstrate OCP-OCS vision toward open OCS standard with Open HW and SW capability.
  - Participants: OCS Vendors and Data Center Research Operator
  - OFC-2027 Demo
    - Showcase multi-vendor interop proof of concept (poc) using OCP-OCS Standard YANG being managed by centralize SDN controller without any custom vendor YANG translator for HW resiliency application.

Call to Action

- **Join our OFC Demo!**
  - Experience interoperable multi-vendor OCSs in action and see the real-world capabilities
- **Participate in Discussions at OCS subproject!**
  - Engage in ongoing OCP and IOWN-GF use case discussions to shape the future of optical AI interconnects
  - Further idea about future PoC or use-cases
- **Contribute your software!**
  - Ongoing demo uses proprietary controller → Need for **Open-Source** OCS SDN Controller
  - Future topic: Integration with AI orchestrator and OCS SDN controller

- OCP OCS Subproject: https://www.opencompute.org/community/optical-circuit-switching
- Mailing List: https://ocp-all.groups.io/g/ocp-optical-circuit-switching


Thank
You!
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San Jose, California

Appendix: scaling challenges

Challenge 1 - The Bandwidth & Power Wall

Electrical Interconnect Scaling Limits

	Electrical lanes >200Gbps/ch	Optical Pluggable TRx >200Gbps/ch
Distance	< 1.5m	> 10m
XPUs	< 100	> 1,000
Energy Efficiency	~2 pJ/bit (DAC) 2~5 pJ/bit (ACC, AEC)	10~15 pJ/bit

EPS Power Scaling Toward 1 kW

Year	Switching Capacity	Typical Power (per ASIC)
2018	12.8 Tbps	350–400 W
2020	25.6 Tbps	450–500 W
2022	51.2 Tbps	700–800 W
2024	102.4 Tbps	900–1000 W

Challenge 2 - Resiliency Bottlenecks

MTBF vs. # of GPUs

